

Name: Team Timbuktu

Date: 2/15/25

Element F: Consideration of design viability

The proposed design for the fence hanger needs to extend over an 8-foot tall fence to allow zookeepers easy access to certain toys over the fence. The structure must extend 5-6 feet over it to give animals enough room to interact with the toys without feeling cramped. For stability, the structure must also be 2-3 feet into the ground to withstand animal interactions and weather conditions. Using concrete to secure the buried portion will make it even more stable during strong winds or when animals lean against it.

The vertical post will be connected to a horizontal post that extends outward 6 feet from the attachment point. A metal bracket should reinforce this connection point to prevent weakening over time. The first 2 feet of the horizontal will be behind the fence on the side of the vertical post, while the remaining distance will extend past the fence and over the enclosure. This balanced design prevents the structure from tipping forward when weight is applied to the extended end.

Along the bottom of the horizontal post, a 1- $\frac{5}{8}$ " metal strut will hang securely using heavy-duty clamps. Inside the strut will rest a trolley designed for use inside said struts, with smooth-rolling wheels to allow easy movement. A ratio pulley will hang from the trolley, and the toy will hang from that at a height that's comfortable for the animals to reach. The rope used in the ratio pulley will go to the vertical post and attach at the bottom to a cleat that will hold it in place firmly. Adding a secondary safety knot below the cleat would prevent accidents if the rope slips.

A second rope will also attach to the trolley and go through pulleys mounted to the 4x4s, allowing the zookeepers to adjust the location of the trolley as they see fit. This rope system should include smooth-running ball-bearing pulleys to make operation easier for the keepers. The design also allows the trolley to pass over the fence and offload the

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smaller toys from the same side as the ropes, creating a convenient workflow for staff members. Marking the ropes with different colors could help keepers quickly identify which rope controls which function.

The materials were chosen for strength and cost effectiveness while ensuring durability. The 4x4s have sufficient strength while being one of the cheaper options for structural materials, and using pressure-treated wood will help them last longer outdoors. The 1- $\frac{5}{8}$ " strut and trolley were chosen because we have access to the strut and can easily find a trolley for the given purpose. These components should be weather-resistant to prevent rust and corrosion over time. The height of the post was carefully calculated because it has to extend at least 2 feet above the 8-foot fence for some of the smaller toys to pass over the fence and 2-3 feet down to seat it in concrete and make it stable against various forces.

The horizontal extension was specifically designed because the vertical has to sit 2-3 feet away from the fence to allow for easy operation by staff members. It has to stick out 3-4 feet from the fence to allow the Elands to play with the toys without hitting the fence or feeling restricted in their movements. The rope was chosen because a group member has nearly 40 feet of it at their house and, after some simple testing, was determined to be able to hold the weight of the toys. However, it would be wise to test the rope with weights exceeding the expected toy weight to ensure an adequate safety margin. Regular inspection of all ropes and connections should be part of the maintenance routine to catch any wear before it becomes dangerous.